INAIO 2025 -Stage 1

Instructions

- Exam Timings: 25th Jan 10:00 - 11:30 am IST. Test will close at 11:30 am IST.
- Remember to save your answers. Only the final saved answers will be counted towards your score.
- Ensure you have good internet connectivity and are alone in a quiet place.
- Students are expected to work on the problems individually without assistance from anyone else.
- Marks Breakup:
  - Total: 80 marks + up to 20 bonus marks:
  - 10 questions worth 2 points each (approx. 2 minutes per question)
  - 4 multi-part questions worth 15 points each (approx. 15 minutes per question)
  - Bonus Marks: To incentivise original thinking and discretion in using external tools, bonus marks would be awarded for questions where tools such as ChatGPT tend to make errors. The list of the problems for which bonus marks will be awarded will be published the week after the exam.
- Attempt all questions. There are no optional questions.
- There are no negative marks.
- All questions will have answers that are either one of the few discrete options, or an integer within the range of 0 to 99. In case of numeric answers, round the final answer downward to the nearest integer.
- Students can go back and forth between the sections to update their answers before the final submission.
- Students are permitted to use the free version of ChatGPT, Google, Python notebooks, and online calculators during the test, but do not need to use these tools. Note that ChatGPT often tends to give erroneous answers. It might often be faster and more accurate to solve the problems directly.
- For each problem, we also provide a text box for you to briefly describe your solution approach. These can be rough cryptic notes, and will not be used for Stage 1 scoring. It is optional and primarily meant to gain insights into how students are approaching the problems. This could help with later stages of the selection process. Do mention the use of tools. Fluency and discretion in using the allowed tools will be viewed as positive.
- Please be aware that multiple simultaneous accesses to the test from a single registered Google account will lead to disqualification.
- Results of the test will be communicated through email by Feb 5.
- In case you encounter challenges or have questions, please contact: `inaio@acmindia.org`.

Short Problems

Wizardly Wason Safety Check

You are the Hogwarts Charms professor, ensuring students follow the below safety rule during spell practice:

> *If a student is casting either Expelliarmus or Stupefy, then they must either have Professor McGonagall's or Dumbledore's permission.*

You observe 8 students practicing spells through Hogwarts Observation Cards. Each card represents a student. One side of the card shows the charm the student is casting (e.g., Expelliarmus, Stupefy, Lumos, etc.) while the other side shows information on the permissions provided. Note that students can cast at most one spell at a time.

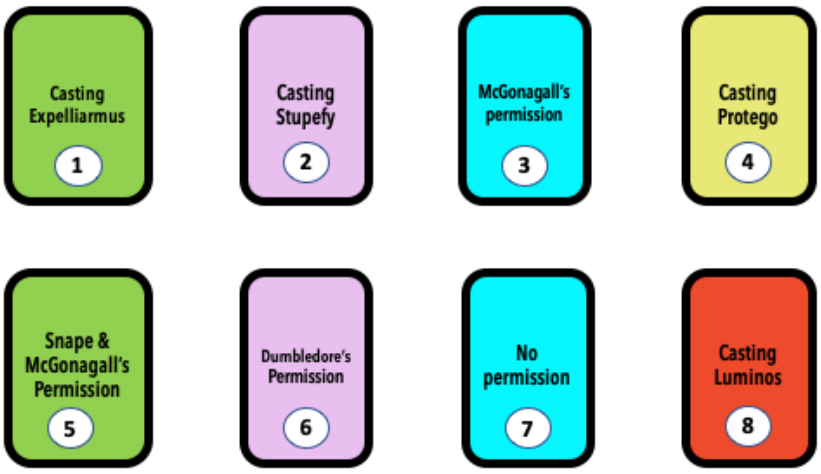


Figure 1: Hogwarts Charms Observation Cards

Question: Which card(s) must you turn over to check if there are any safety rule violations?

Options:

1. $[1, 2, 3, 5, 6]$
2. $[1, 2]$
3. $[1, 2, 7]$
4. $[3, 4, 5, 6, 8]$
5. $[3, 5, 6]$
6. $[4, 7, 8]$

RoboChef's Taste Test

You have built "RoboChef," which can taste food and list its ingredients. RoboChef is now presented with a simple dish containing a single condiment ingredient. It is known that the dish can be:

- Salt, with a 60% chance
- Sugar, with a 30% chance
- Pepper, with a 10% chance

RoboChef is equipped with a salt sensor. When a taste test is performed, the sensor behaves as follows:

- If the dish has salt, the sensor claims "Salt found" 90% of the time.
- If the dish does **not** have salt, the sensor claims "Salt found" 30% of the time.

Question: RoboChef performs a taste test, and the sensor displays "Salt found." Let p be the probabilitythat the condiment is either Salt or Pepper? Report $\lfloor 100p \rfloor$, the largest integer less than or equal to $100p$.

Perfectly Priced Park Ticket

You are a finance consultant for an amusement park owner. Each day, the park can host up to 100 visitors, but attendance depends on the ticket price—higher prices mean fewer visitors, and lower prices mean more. Specifically, there are 100 visitors when it is free and for every 1 rupee increase in the ticket price, the park sees one fewer visitor. Running the park has a fixed cost of 500 rupees daily. Additionally, every visitor adds 10 rupees to daily costs for maintenance and services.

Your goal is to set a ticket price that maximizes profit while also meeting these conditions:

- At least 20 people must visit each day to keep the park "popular."
- The ticket price must be at least 50 rupees, as the owner refuses to go lower.

Question: What entry fee should you suggest to maximize profit while satisfying the conditions?

Mystery Balls and Balances

You have four balls labeled A, B, C, and D. Among these:

- One of A or B weighs 100 g, and the other weighs 101 g.
- One of C or D weighs 200 g, and the other weighs 201 g.

You do not know which is which within each pair (A, B) and (C, D). You also have three identical beam balances, but only two of them work perfectly. The third is unreliable and only occasionally gives correct readings. You do not know which two balances are perfect.



Figure 2: Beam Balance Setup for Weighing Balls

Questions:

a. What is the minimal number of weighings required to determine the exact weights of A and B using the three beam balances?

b. If one perfect beam balance is taken away (leaving you with one perfect and one unreliable), how many weighings are needed to determine the exact weights of A and B?

c. What is the minimal number of weighings required to determine the exact weights of A, B, C, and D using the three beam balances?

Options:

1. (a) 3, (b) 6, (c) 6
2. (a) 3, (b) Not Possible, (c) 6
3. (a) 2, (b) 6, (c) 6
4. (a) 3, (b) Not Possible, (c) 7
5. (a) 3, (b) Not Possible, (c) 4
6. (a) 3, (b) Not Possible, (c) 5

Fraud Filters

Background: Binary Classification Metrics

Binary classifiers typically provide a score indicating how likely each instance is to be positive. A threshold (operating point) is chosen to decide whether to label an instance as positive or negative. The choice of this threshold depends on application constraints and can be guided by metrics such as precision, recall, accuracy, or coverage, particularly in cases of class imbalance.

Confusion Matrix:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

1. **Precision:** The proportion of positive predictions that are correct.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

2. **Recall:** The proportion of actual positives that are correctly predicted.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

3. **Accuracy:** The proportion of all correct predictions (both positive and negative) out of the total predictions.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}}$$

4. **Coverage:** The total number of predictions made for the positive class.

$$\text{Coverage} = \text{True Positives} + \text{False Positives}$$

An e-commerce company employs a binary classifier to identify potentially fraudulent transactions, which are then forwarded to a human expert team for review. It is known that 1 in 10 transactions is fraudulent. However, the human expert team has the capacity to review only 1000 transactions per day.

Question: In this scenario, which criterion should be used to determine the appropriate score threshold (operating point) for the classifier?

Options:

1. Maximize Accuracy
2. Maximize Coverage
3. Maximize Recall
4. Maximize Recall at a Precision Bound
5. Maximize Precision at a Coverage Bound
6. Maximize Precision at a Recall Bound

Revenue RollerCoaster Function

The plot below shows the revenue $R(t)$ of Apple AirPods over time (t, in months). The revenue is typically influenced by several factors including baseline behavior, seasonality, decay, and occasional spikes due to promotions.

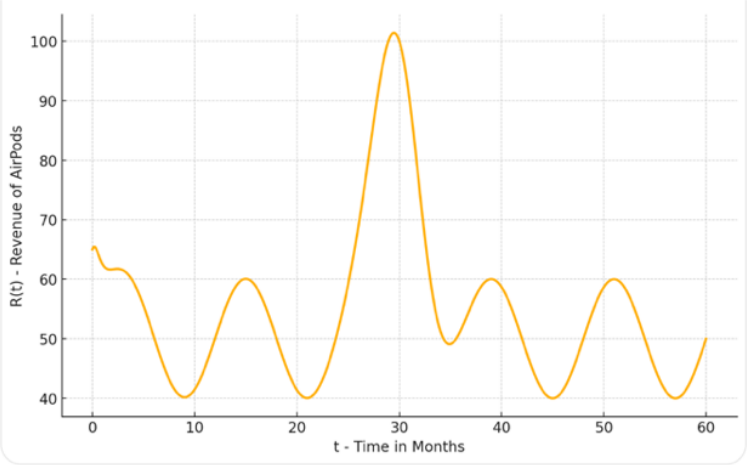


Figure 3: AirPods Sales Revenue Over Time

Question: Based on the plot and description, identify the function $R(t)$ that best models the revenue of AirPods.

Options:

1. $R(t) = 50 + 20 \cdot \cos\left(\frac{2\pi t}{12}\right) + \frac{30}{1+t} + 10 \cdot e^{-0.1(t-30)^2}$
2. $R(t) = 50 + 10 \cdot \sin\left(\frac{2\pi t}{12}\right) + \frac{15}{1+t^2} + 50 \cdot e^{-0.1(t-30)^2}$
3. $R(t) = 50 + 15 \cdot \sin\left(\frac{\pi t}{6}\right) + 5 \cdot \ln(t+1)$
4. $R(t) = 50 + 10 \cdot \sin\left(\frac{2\pi t}{12}\right) + 20 \cdot e^{-t/20}$
5. $R(t) = 50 + 10 \cdot \sin\left(\frac{2\pi t}{12}\right) + \frac{15}{1+t^2}$
6. $R(t) = 50 + 5 \cdot \ln(t+1) + \frac{10}{t+1} + 30 \cdot e^{-0.1(t-30)^2}$

Study-Play Balance for Success

You are analyzing the total performance score y of a high school student at the end of the semester. The score is influenced by:

- Hours of Study per Week (x_1): The average number of hours the student studies each week.
- Participation in Extracurricular Activities (x_2): A positive factor (1 to 10) representing the student's involvement in sports, clubs, or volunteering.

The relationship between the total score (y), hours of study (x_1), and extracurricular participation (x_2) is approximately linear:

$$y \simeq a_1 \cdot x_1 + a_2 \cdot x_2 + y_0,$$

where a_1 and a_2 are integers.

Table 1: Total Score (y) vs. Hours of Study (x_1) for $x_2 = 5$

Hours of Study	**Total Score**
5	276
10	324
15	376
20	423

Table 2: Total Score (y) vs. Extracurricular Participation (x_2) for $x_1 = 5$

$\mathbf{x}_2$	**y**
3	244
5	276
7	303
9	334

Question: What is the student's total score if they study 12 hours per week and have an extracurricular participation score of 8?

Options:

1. 240
2. 390
3. 415
4. 440
5. 340
6. 485

Party Planning Puzzle

Your class of 30 students has raised 3000 INR for hosting a party. You are tasked with ordering pizzas, fresh juice, and soda bottles while meeting these requirements:

1. Each student will have **either fresh juice or soda**, but not both.
2. Each student will receive **one-sixth** of a pizza, and **2 pizzas** are reserved as a thank-you gift for the cleaning staff.
3. The total cost of pizzas, fresh juice, and soda must not exceed the budget.

Costs:

- A pizza costs 300 INR.
- A fresh juice bottle costs 50 INR.
- A soda bottle costs 20 INR.

Question: How many total items (pizzas, fresh juice bottles, and soda bottles) should you order to satisfy all the constraints?

Water Balloon Toss

At a school festival, there is a water balloon toss game. Each participant gets up to 8 balloons to throw at a set of 10 cans. Here are the rules:

- A participant must stop throwing as soon as they have 2 consecutive misses.
- Each balloon has a 50% chance of hitting a can.
- Participants win if they hit at least 5 cans.

Question: Let p be the probability that a participant will win the game by knocking down at least 5 cans. Report $\lfloor 100p \rfloor$, the largest integer less than or equal to $100p$.

Similarity Sleuth

You are a data scientist trying to reverse engineer a similarity function used in some blackbox software that you cannot read. To do this, you construct four vectors A, B, C, and D, as shown in Table 1, and observe the outputs of the pairwise similarity function, as displayed in Table 2.

Vector	Feature 1	Feature 2	Feature 3
A	1	2	0
B	0	3	4
C	2	1	3
D	3	0	1

Table 1: Four vectors with their corresponding features.

	A	**B**	**C**	**D**
A	1.00	0.53	0.47	0.42
B	0.53	1.00	0.80	0.25
C	0.47	0.80	1.00	0.76
D	0.42	0.25	0.76	1.00

Table 2: Pairwise similarity values between the vectors. Note: The actual test had an error in the matrix and hence, are discounting this question while scoring.

Question: From other sources, you know that the similarity function is one of the options listed below. Can you identify the similarity function used in the blackbox software?

Options:

1. **Cosine Similarity:**
$$\text{Similarity}(u, v) = \frac{\sum u_i v_i}{\sqrt{\sum u_i^2} \cdot \sqrt{\sum v_i^2}}$$
2. **Dot Product:**
$$\text{Similarity}(u, v) = \sum u_i v_i$$
3. **Negative L1 Distance with Offset:**
$$\text{Similarity}(u, v) = C - \sum |u_i - v_i|$$
where C is a positive constant.
4. **Exponential of Negative L2 Squared Distance:**
$$\text{Similarity}(u, v) = \exp\left(-\sum (u_i - v_i)^2\right)$$
5. **Mahalanobis Similarity:**
$$\text{Similarity}(u, v) = 1 - \sqrt{(u - v)^T S^{-1} (u - v)}$$
where S is the covariance matrix of the data, defined as:
$$S = \frac{1}{n-1} \sum_{i=1}^{n} (u_i - \bar{u})(v_i - \bar{v})^T,$$
with $\bar{u}$ and $\bar{v}$ being the means of the vectors.

6. **Pearson's Correlation:**

$$\text{Similarity}(u, v) = \frac{\sum(u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum(u_i - \bar{u})^2} \cdot \sqrt{\sum(v_i - \bar{v})^2}}$$

where $\bar{u}$ and $\bar{v}$ are the means of u and v, respectively.

Long Problems

Alumni Donor Classifier

Background: Multi-class Classification

In a multi-class classification problem, the goal is to assign instances to one of several possible classes. The performance of the classifier is evaluated using the following metrics.

1. Confusion Matrix The confusion matrix for a 3-class problem is structured as:

$$\begin{bmatrix} TP_{C1} & F_{C1,C2} & F_{C1,C3} \\ F_{C2,C1} & TP_{C2} & F_{C2,C3} \\ F_{C3,C1} & F_{C3,C2} & TP_{C3} \end{bmatrix}$$

where:

- TP_{C_i}: True positives for class C_i (instances correctly classified as C_i).
- F_{C_i,C_j}: Instances of class C_j incorrectly classified as C_i.

2. Precision (P_i) The proportion of correctly classified instances for class C_i out of all instances predicted as C_i:

$$P_i = \frac{TP_{C_i}}{TP_{C_i} + \sum_{j \neq i} F_{C_i,C_j}}$$

3. Recall (R_i) The proportion of correctly classified instances for class C_i out of all actual instances of C_i:

$$R_i = \frac{TP_{C_i}}{TP_{C_i} + \sum_{j \neq i} F_{C_j,C_i}}$$

4. ROC Curve The Receiver Operating Characteristic (ROC) curve is a graphical representation of the tradeoff between the true positive rate (TPR) and the false positive rate (FPR) at various thresholds. For binary classification:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

The ROC curve is extended for multi-class problems by focusing on one class (positive) against the rest (negative). The various terms are based on the positive and negative class definitions.

A school classifies its alumni into three classes based on various criteria. The classes and their actual distribution is given below.

- Top donors (C_1): 25%
- Mid-level donors (C_2): 50%
- Minimal donors (C_3): 25%

Two classifiers A and B are evaluated for their performance on the classification task and below are their precision and recall metrics specific to each class.

Classifier A Metrics:

Class	Precision	Recall
C_1	0.85	0.70
C_2	0.80	0.85
C_3	0.75	0.80

Classifier B Metrics:

Class	Precision	Recall
C_1	0.90	0.65
C_2	0.78	0.82
C_3	0.80	0.85

Questions:

Round downwards to nearest integer for numeric values.

Q1: Confusion Matrix Calculation

Compute the confusion matrices for Classifier A and Classifier B using the class ratios and metrics. In the questions below, fractions are with respect to the entire set of instances. For example, if True Positives (for some choice of positive class) form half of all the instances, then True Positive fraction (in %) would be 50 for that class.

a. What is the True Positive fraction (in %) for Classifier A with C_1 as the positive class and rest as negative ? [1 pt]

b. What is the False Positive fraction (in %) for C_2 (i.e., fraction of actual C_1 and C_3 instances mistaken for C_2) for Classifier B? [1 pt]

c. What is the False Negative fraction (in %) C_3 (i.e, fraction of instances predicted as C_3 but actually C_1 or C_2) for Classifier A? [1 pt]

Q2: New Classifier (AB) Definition

A new classifier (AB) is defined as follows:

- For men, use classifier A.
- For women, use classifier B.

If it is known that the men:women ratio among the alumni is 3:1, compute the confusion matrix for the new classifier AB.

a. What is the precision (in %) of Classifier AB for C_1? [2 pts]

b. What is the recall (in %) of Classifier AB for C_2 ? [2 pts]

Q3: ROC and Class Distribution

The ROC curve for class C_3 (as the positive class) vs. the rest is provided for the classifier AB.

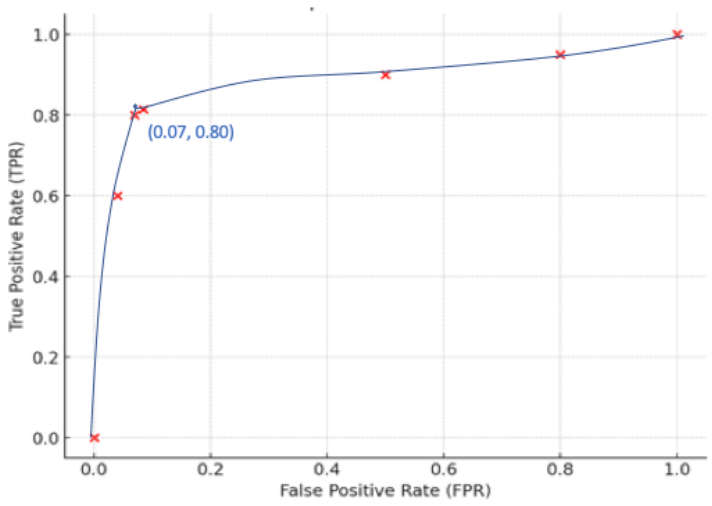


Figure 4: ROC for Class C_3 from Classifier AB

Based on the given ROC curve, answer the below questions.

a. What is the accuracy (in %) of Classifier AB for recall of 0.8 with respect to C_3 ? [2 pts]

b. If the class distribution changes to a distribution $(50\%C_1, 25\%C_2, 25\%C3)$, how does the ROC curve change? [2 pts]

Options:

1. The curve shifts upward.
2. The curve shifts downward.
3. The curve remains unchanged.
4. The vertical and horizontal axes stretch in the ratio $2 : 0.75$ respectively.
5. The vertical and horizontal axes stretch in the ratio $0.75 : 2$ respectively.
6. None of the above

c. For the above setting, if the positive class (C_1) and negative classes (C_2 and C_3) are flipped, how does the ROC curve change? [2 pts]

Options:

1. The curve remains the same.
2. The curve compresses vertically.
3. The curve compresses horizontally.

4. The curve flips across the diagonal.
5. The curve shifts upwards
6. The curve shifts downwards

Q4: Mis-classification Cost

Based on class predictions and the actuals, the school incurs the following gain/losses (in 1000 INR) per each alumnus.

Classification Cost Matrix

Actual (Rows)/ Predicted (Cols)	C_1	C_2	C_3
C_1	50	-10	-40
C_2	-20	30	-15
C_3	-50	-20	10

Compute the average gain (in 1000 INR) for Classifier AB given its confusion matrix. [2 pts]

Matrix Magic for Image Processing

Background: Image Transformation via Convolution

1. **Matrix (Image Representation):** A grayscale image can be represented as a matrix where each value corresponds to the pixel intensity:
   - **0** represents black.
   - **255** represents white.
   - Values in between represent shades of gray.
2. **Kernel (Filter):** A smaller matrix used to apply transformations to an image, such as blurring, sharpening, or edge detection.

   [width=0.8]figures/sharpen.png
3. **Convolution:** Convolution is a mathematical operation that combines two functions to produce a third function expressing how the shape of one function is modified by another. In image processing, for a 2D image $I(x, y)$ and a kernel $K(i, j)$, the convolution $O(x, y)$ is given by:

$$O(x, y) = \sum_i \sum_j K(i, j) \cdot I(x + i, y + j)$$

   where:
   - $O(x, y)$: The output value at position (x, y).
   - $I(x, y)$: The input pixel value at position (x, y).
   - $K(i, j)$: The kernel value at position (i, j).

Background: Image Transformation via Convolution (Cont.)

4. **Padding:** Padding is the process of adding extra rows and columns around the borders of an image or matrix before applying convolution. This ensures that the kernel can be applied to edge pixels without reducing the dimensions of the output.

   [width=0.7]figures/padding.png
5. **Stride:** Stride refers to the step size by which the convolution kernel moves across the image or matrix during the convolution operation. A stride of 1 means the kernel shifts one pixel at a time, whereas a stride of 2 means the kernel shifts two pixels at a time.

   [width=0.7]figures/stride.png

Background: How to Perform Convolution?

Steps to implement Convolution Formula:

1. Place the kernel over a sub-region of the image matrix (top-left corner initially).
2. Multiply each element of the kernel with the corresponding element of the image matrix.
3. Add the results to get a single value, which becomes the output pixel for that region.
4. Slide the kernel across the image, repeating steps 1–3, until the entire image is processed.

Example: Convolution Process

Image Matrix (3x3):

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Kernel (2x2):

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Final output matrix will be:

$$\begin{bmatrix} 6 & 8 \\ 12 & 14 \end{bmatrix}$$

Note: The convolution performed in the above example had stride = **1** and **did not** use any padding.

Questions

Q1: Performing Convolution [3 pts]

Using the following **image matrix** and **kernel matrix**, perform convolution and calculate the the output matrix. **Note:** Do not perform padding for the input matrix and assume stride = 1.

Image Matrix (5x5):

$$\begin{bmatrix} 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 7 & 8 & 9 \\ 8 & 8 & 8 & 8 & 8 \\ 7 & 6 & 5 & 4 & 3 \\ 3 & 2 & 1 & 0 & 0 \end{bmatrix}$$

Kernel (3x3):

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

Report the **score** of the output matrix A defined below as sum over values of "even" cells:

$$\textbf{score}(A) = \sum_{i,j| \quad i+j \text{ is even}} a_{ij}.$$

Q2: Stride and Padding [2 pts]

Suppose you have an input image of size 98 x 100. You apply the convolution operation on this image using a kernel of size 4 x 4. Assume **no padding** and **stride** = 4. Find the dimensions of output matrix and report $n + m$ if n and m are the dimensions of the output matrix?

Q3: Blurring [2 pts]

Blurring reduces the sharpness of an image by smoothing out fine details. The result is a softer appearance, where individual pixels blend more closely with their neighbors. Based on this information, which kernel would most effectively blur the image?

Options:

1. $$1/9 * \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

2. $$\begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{bmatrix}$$

3. $$9 * \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

4. $$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Q4: Brightening [2 pts]

Brightening makes the image lighter resulting in a more luminous or whiter image. Based on this information, which kernel is best suited to brighten the image?

Options:

1. $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

2. $$\begin{bmatrix} -1 & 0 & -1 \\ 0 & 5 & 0 \\ -1 & 0 & -1 \end{bmatrix}$$

3. $$\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

4. $$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q5: Sharpening [2 pts]

Sharpening increases contrast around edges, making boundaries and details in the image more pronounced and clear. It effectively emphasizes fine details in the image while suppressing smooth areas. Based on this information, which kernel would be best suited for sharpening an image?

Options:

1. $$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

2. $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

3. $$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

4. $$\begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{bmatrix}$$

Q6: Edge Detection [2 pts]

Edge detection identifies areas in the image where there are abrupt changes in pixel intensity, such as the boundary between two objects. This operation highlights these transitions, making the edges stand out. Based on this information, which kernel is best suited for edge detection?

Options:

1. $$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

2. $$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

3. $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

4. $$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Q7: No Change [2 pts]

Assuming the original image is appropriately padded with 0's before applying convolution, which of the following kernels will preserve the original image as is?

Options:

1. $$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. $$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3. $$\begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{bmatrix}$$

4. $$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

From Words to Vectors - Teaching Language to Machines

Background: Word Embeddings

Word embeddings are a foundational technique in natural language processing (NLP), where words are converted into vectors in a multi-dimensional space. This method captures semantic relationships by representing words as numerical vectors that reflect their different semantic features.

For a basic understanding, consider these 2-dimensional word embeddings:

$$\begin{aligned} \text{Happy:} &\quad \begin{bmatrix} -5.2 & 2.3 \end{bmatrix} \\ \text{Joyful:} &\quad \begin{bmatrix} -4.8 & 2.6 \end{bmatrix} \\ \text{Sad:} &\quad \begin{bmatrix} -5.3 & -2.3 \end{bmatrix} \\ \text{Tasty:} &\quad \begin{bmatrix} 5.9 & 2.9 \end{bmatrix} \\ \text{Tasteless:} &\quad \begin{bmatrix} 5.7 & -1.9 \end{bmatrix} \end{aligned}$$

Here, "Joyful" and "Happy" have very similar embeddings, indicating their close semantic relationship. However, the vector for "Sad" shares a similar first dimension with "Happy" and "Joyful" but differs significantly in the second dimension, reflecting the opposite emotional content. Similarly, "Tasty" and "Happy" share proximity in the second dimension, suggesting a positive sentiment, but differ in the first dimension, which likely represents different domains (taste vs. mood). In general, each dimension of the embeddings pertains to distinct semantic aspect.

Questions:

Q1: Predicting Missing Word Embeddings [2 pts]

You have computed the 2-dimensional word embeddings for three words: "king", "queen", "man".

- King: $\begin{bmatrix} -2 & -2 \end{bmatrix}$
- Queen: $\begin{bmatrix} -2 & 2 \end{bmatrix}$
- Man: $\begin{bmatrix} 2 & -2 \end{bmatrix}$

Based on the embeddings above, predict the most likely embedding for "woman".

Options:

1. $\begin{bmatrix} 2 & 2 \end{bmatrix}$
2. $\begin{bmatrix} -2 & 2 \end{bmatrix}$
3. $\begin{bmatrix} 2 & -2 \end{bmatrix}$
4. $\begin{bmatrix} 0 & 2 \end{bmatrix}$
5. $\begin{bmatrix} -2 & -2 \end{bmatrix}$
6. $\begin{bmatrix} 1 & 2 \end{bmatrix}$

Q2: Semantic Difference between Word Embeddings [2 pts]

Using your answer from Q1, determine what semantic concept the vector calculated from subtracting the embedding of "woman" from "queen" (Queen - Woman) most likely represents.

Options:

1. Man
2. King
3. Gender
4. Royalty
5. Female
6. Job

Q3: Maximizing Sentence Incoherence [2+5+4 pts]

Coherence of sentences is typically measured by the semantic relatedness and logical consistency of words within a sentence. A common method to quantify coherence involves calculating the semantic "distance" between adjacent words using their word embeddings. This distance can be expressed through a "loss" metric, with lower values indicating higher coherence.

In the fictional city of Siltover, sentence incoherence is quantified using the Manhattan distance between word embeddings **a** and **b** which is defined as:

$$\text{Manhattan Distance} = \sum |a_i - b_i|$$

where a_i and b_i are the components of vectors **a** and **b**, respectively.

For example, consider the sentence "Arcate Siltover RexTech" with the following embeddings:

- Arcate: $\mathbf{a} = \begin{bmatrix} 10 & -2 \end{bmatrix}$
- Siltover: $\mathbf{b} = \begin{bmatrix} 9 & 8 \end{bmatrix}$
- RexTech: $\mathbf{c} = \begin{bmatrix} 3 & -2 \end{bmatrix}$

The loss for this sentence, calculated as the sum of the Manhattan distances between consecutive embeddings, is $|10 - 9| + | - 2 - 8| = 11$ for Arcate to Siltover and $|9 - 3| + |8 + 2| = 16$ for Siltover to RexTech, resulting in a total loss of 27.

While maximizing coherence is typical, Siltover's citizens seek novelty in linguistic expression. They have launched the "King of Gibberish" contest, challenging participants to craft sentences with the highest possible incoherence using the following word embeddings:

- RexTech: $\begin{bmatrix} 3 & -2 \end{bmatrix}$
- Chimmer: $\begin{bmatrix} -15 & -3 \end{bmatrix}$
- Xaun: $\begin{bmatrix} -7 & 8 \end{bmatrix}$
- Arcate: $\begin{bmatrix} 10 & -2 \end{bmatrix}$
- Siltover: $\begin{bmatrix} 9 & 8 \end{bmatrix}$

As part of the "King of Gibberish" contest, you need to find creative chaotic sentences

a. First, starting with "RexTech," identify the next word that maximizes loss for a two-word sentence. Report this maximal loss value. [2 pts]

b. Starting with "RexTech", make a four-word sentence by adding three more words that achieve the highest total loss. Report this maximal loss value. [5 pts]

c. For the final level of the "Gibberish" challenge, a new incoherence loss ("All Pair Manhattan Distance loss") is introduced. This loss is calculated as the sum of the distances between all possible pairs of word embeddings in a sentence. Construct a three-word sentence starting with "RexTech" by adding two more words that maximizes this loss and determine the maximum possible loss value. [4 pts]

PageRank: Mapping User Influence

Background: PageRank

PageRank is an algorithm originally designed to rank web pages by analyzing the hyperlink structure of the web. Each page is assigned a score representing its relative importance, which is iteratively computed based on links from other pages. Due to its simplicity and effectiveness, PageRank is used widely for ranking network entities and measuring influence across various domains.

Weighted PageRank is a generalization of the original PageRank algorithm that introduces weights to outgoing edges, allowing for unequal influence allocation. The iterative formula for the PageRank of node i at iteration k is:

$$PR^{(k)}(i) = \frac{(1-d)}{N} + d \cdot \sum_{j \in \text{In}(i)} \frac{PR^{(k-1)}(j) \cdot W[j][i]}{S[j]}$$

where:

- N: Total number of nodes.
- d: Damping factor, representing the chance of following a link versus jumping to a random node.
- $W[j][i]$: Weight assigned from node j to i.
- $S[j] = \sum_k W[j][k]$: Total outgoing weight from node j.

To better understand the PageRank algorithm, it is useful to consider the **transition matrix** T, which represents the probabilities of moving from one node to another in a network, incorporating both the structure of the graph and the damping factor. It is defined as:

$$T[i][j] = d \cdot \frac{W[j][i]}{S[j]} + \frac{1-d}{N}$$

Using the transition matrix T, the PageRank update step becomes

$$PR^{(k)} = T \cdot PR^{(k-1)}$$

so that the PageRank after the kth iteration can be recursively derived as

$$PR^{(k)} = T^k \cdot PR^{(0)}$$

where $PR^{(0)}$ is the initial PageRank vector, typically initialized as $PR^{(0)}[i] = \frac{1}{N}$.

Convergence: For finite well-behaved graphs, as $k \to \infty$, the PageRank computation converges to a steady-state solution, i.e., change in values between iterations is negligible. The steady-state PageRank corresponds to the **principal eigenvector** $\mathbf{x} \in R^N$ of T, which satisfies

$$T\mathbf{x} = \lambda\mathbf{x},$$

where $\lambda = 1$ is the **largest eigenvalue** (in magnitude) of T and $\mathbf{x} \neq \mathbf{0}$.

In practice, the actual convergence depends on a variety of criteria such as the damping factor d, the graph structure, PageRank initialization, and presence of special graph structures such as sink components, and cycles.

A social network platform is analyzing user influence through a Weighted PageRank model. The network is represented as a directed graph with weighted connections. Users assign weights to their outgoing connections based on how influential they perceive others to be as shown in Figure 5. The weight matrix W for a network of 6 users (A, B, C, D, E, F) is as follows, where $W[j][i]$ represents the weight from node i to node j:

$$W = \begin{bmatrix} 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

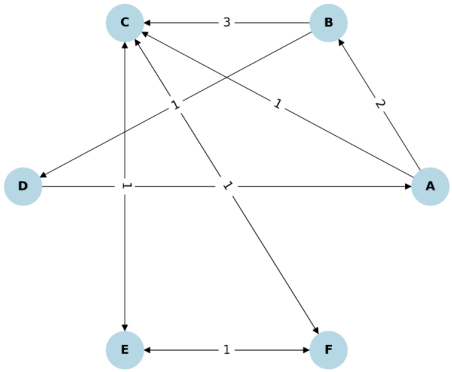


Figure 5: Directed graph showing users A, B, C, D, E, F.

Q1: Applying PageRank

a. Choosing the damping factor d to be 0.9, construct the transition matrix T using the given weight matrix W. Report $\lfloor 100s \rfloor$ where s is the sum of the diagonal elements of T? [2 pts]

b. Using the standard initialization $PR^{(0)} = \frac{1}{N}$ as the initial PageRank for all users, compute the PageRank r of user C after the first iteration. Report $\lfloor 100r \rfloor$. [1 pt]

c. After the first iteration, identify the most influential user, i.e., one with the highest PageRank value. [1 pt]

Options:

1. User A
2. User B
3. User C
4. User D
5. User E
6. User F

Q2: Convergence Properties

a. If d is increased from 0.9 to 0.95, how does this affect the algorithm? [2 pts]

Options:

1. Convergence becomes faster.
2. Convergence becomes slower and final values become more sensitive to the link structure (represented by W).
3. Final values become less sensitive to the link structure.
4. Convergence becomes slower, but final values remain unaffected by the link structure.
5. Convergence rate remains unchanged but final values become less sensitive to link structure.
6. Both convergence and final values are unaffected.

b. Based on the convergence properties of PageRank, for which choice of initialization will the PageRank values remain unchanged across iterations? [2 pts]

Options:

1. A uniform distribution, where $PR^{(0)}[i] = \frac{1}{N}$ for all nodes.
2. The principal eigenvector of the transition matrix T.
3. A random distribution where $\sum PR^{(0)}[i] = 1$.
4. Both Options 2 and 3.
5. The inverse of the transition matrix applied to a uniform vector.
6. None of the above.

Q3: Sink Components

We define a *sink component* as a group of nodes in a directed graph that satisfies the following properties:

- Once entered, there is no way to leave the component (i.e., no outgoing edges lead to nodes outside the component).
- All nodes within the component can reach each other through directed paths.

a. What is the largest sink component in the social network user graph? [1 pt]
Options:

1. Users A, B, C
2. Users C, E, F
3. Users A, D
4. Users D, E, F
5. Users C, D, E, F
6. There are no sink components in this user graph.

b. How does the presence of a sink component in a graph affect the behavior of the PageRank algorithm? [2 pts]

Options:

1. The algorithm will not converge due to infinite loops in sink components.
2. Nodes in the sink component will gradually lose importance.
3. Sink components will absorb most of the PageRank, leaving less for nodes outside.
4. The algorithm behaves as if there are two disjoint graphs corresponding to the sink and non-sink component.
5. Sink components only slow convergence
6. None of the above.

Q4: Influence Maximization

a. User A wants to maximize his influence (PageRank) in an existing social network where users already have different levels of influence. Assuming damping factor $d = 1$ and he can modify only his incoming and outgoing links and there is no reciprocity, which strategy would most effectively increase his PageRank? [2 pts]

Options:

1. Only convince more users to follow A, especially those with high PageRank.
2. Only follow as many influential users as possible to become more visible.
3. Only reduce the number of users A follows and concentrate his influence among currently less influential users.
4. Have an equal number of incoming and outgoing links spread across users of different influence levels.
5. Both Options 1 and 3.
6. Both Options 1 and 2.

b. Which of the following is the most salient structural characteristic of a colluding group of users trying to boost their influence? [2 pts]

Options:

1. There is a unique "hub" user followed by all other users in the group.
2. The group forms a clique with bidirectional links between all member pairs.
3. The group minimizes outgoing links to users outside the group.
4. The group minimizes incoming links from users outside the group.
5. Members of group individually minimize outgoing links to all other users.
6. Members of group individually minimize incoming links to all other users.